

See also

[Worker Stays Here After Completing the Job](#)

[Get Job Orders At Home Only](#)

[Home Location](#)

FootPath

FootPath [object]

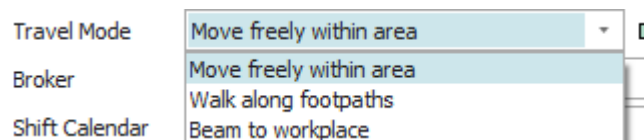
Use the object *FootPath* for modeling a path on which the *Worker* walks from the *WorkerPool* to the *Workplace*.

Image



Description

The *Worker-WorkerPool-Workplace-FootPath*-concept refines the *Broker-Importer-Exporter*-concept. In the **WorkerPool** you can select the **Travel Mode** with which the *Worker* gets to his work station:



- He can **walk freely within the area** of the simulation model, just like in the real world.
- He can be **teleported to the Workplace** you specify.
- He can **walk along the FootPaths** in the simulation model.

As a rule, the *Worker* functions as an **Exporter** with a capacity of 1. If you want to model and simulate walking distances between the *WorkerPool* and work stations or between work stations, use the *FootPath*. *Plant Simulation* animates the *Workers* on the *FootPath* if they move to the stations or in between the stations.

You can link several *FootPaths* with *Connectors* to create a network of *FootPaths*. The *Worker* walks within this network from his *WorkerPool* to the *Workplace*. Provided the *WorkerPool* and the *Workplace* are connected to the same network of *FootPaths*, the *Worker* walks from the *WorkerPool* to the *Workplace* and back on the shortest possible route.

A network of *FootPaths* consists of all *FootPaths* that are joined with *Connectors*, as well as the *WorkerPools* and the *Workplaces* that are joined with one of these *FootPaths* with a *Connector*.

The time it takes the *Worker* to walk from his *WorkerPool* to the *Workplace* depends on his **Speed** and the combined **Lengths** of the *FootPaths* he covers to reach his destination. *Plant Simulation* recognizes on which *FootPaths* the *Worker* walks if he exits a *FootPath* at another end than the one at which he entered it. If the *Worker* does not walk to his destination on a *FootPath* or if he is beamed directly to his destination, he does not consume any time at all.

The *Workers* are created in the **WorkerPool** and they stay there if they do not work and are waiting for a work order.

The **Broker** brokers the *Worker* to the individual work stations. As soon as a *Worker* can provide a service, the *Broker* sends him from his *WorkerPool* to the work station that ordered him:

- If you modeled the work station using one of the material flow objects and the *Workplace* and connected the *WorkerPool* and the *Workplace* with a *FootPath*, *Plant Simulation* animates the *Worker* while he walks on the *FootPath* and while he works on the *Workplace*.

Note

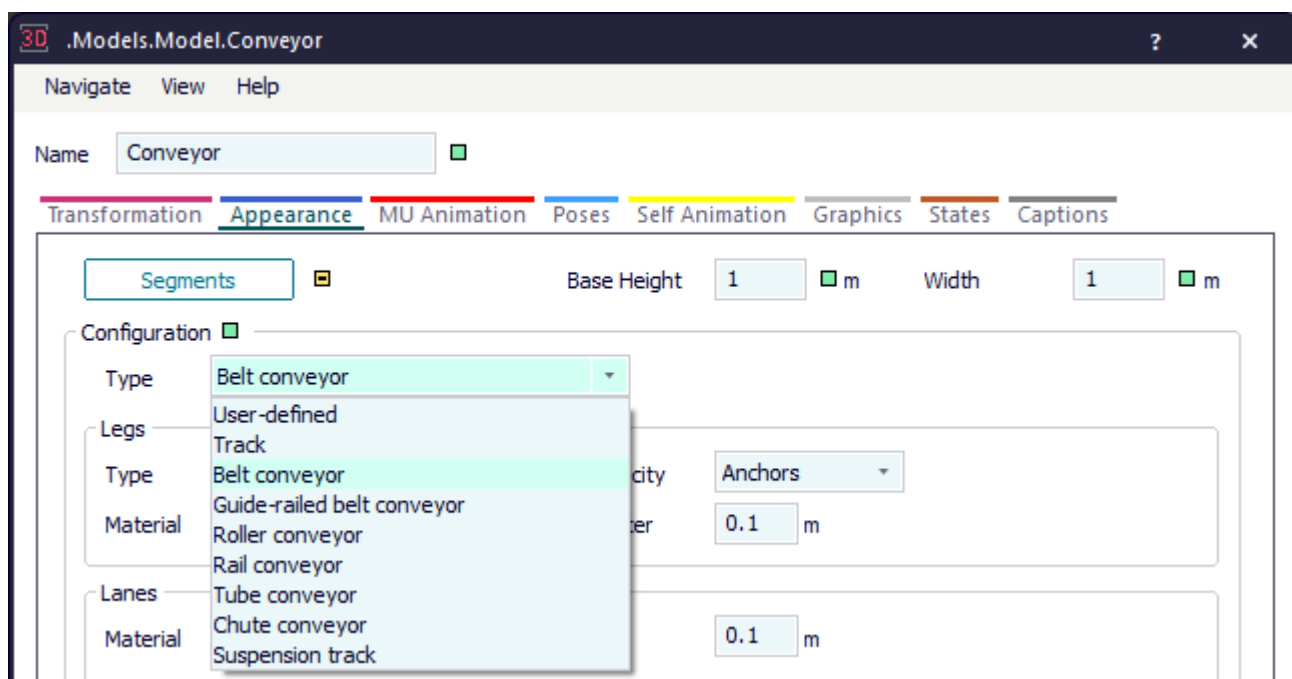
The *Worker* only consumes time if he walks on the *FootPath*.

- If you inserted a *Workplace*, and the *Worker*, who is supposed to work there, cannot reach it via a *FootPath*, *Plant Simulation* beams the *Worker* directly to the *Workplace* and animates him there.
- If you do not insert a *Workplace*, if no *Workplace* exists that supports this service, or if all *Workplaces* that support this service are occupied, *Plant Simulation* beams the *Worker* to the work station itself, where he will perform his task. *Plant Simulation* does not animate him there, but continues to animate him in the *WorkerPool*.

You can insert the *FootPath* into a *Frame*:

- As a curved object, which is the default setting.
- By inserting any sequence of curved segments and straight segments, you can realistically model curved conveyor systems on which the *Transporters* move.

You can select different configurations for the *FootPath* on the tab **Appearance**.



To show a **tooltip** with information about the *FootPath*, hover with the mouse over it.

Show Manipulators

To change the length of the graphic and the anchor points of the object, click **Show Manipulators**  on the **Edit** ribbon tab or press **M** on the keyboard.

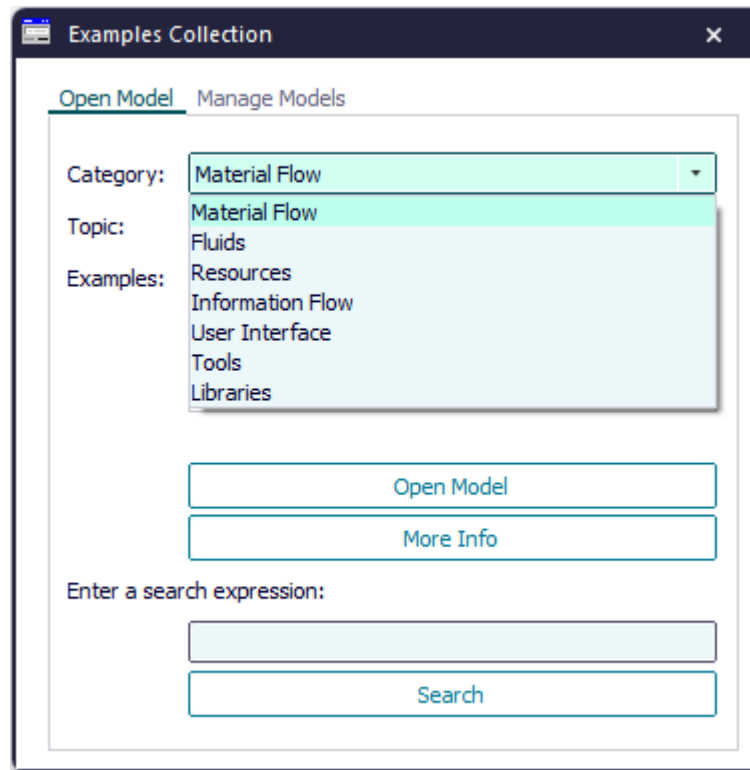
The manipulator at the start and at the end of the length-oriented object is cut off. When you attach an object of the same type to the other object, the two combined halves result in a complete manipulator again. When you move the mouse over a manipulator, *Plant Simulation* shows a *Tooltip*.



Add the Object to the Simulation Model

To add the object *FootPath* to your simulation model, click  **Manage Class Library** > **Basic Objects** > **Resources** > **FootPath** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Work with Length-oriented Objects](#)

[Model Workers and the Jobs They Do](#)

[Worker-WorkerPool-Workplace-FootPath-concept](#)

[Broker-Importer-Exporter-concept](#)

Video on YouTube

https://youtu.be/Rj5eYZu8pl4?si=eMS1tM55tJ4k9_ee&t=420

<https://youtu.be/Rj5eYZu8pl4?si=zfQFRHZgC336hXB7&t=1040>

Dialog Box of the FootPath

Dialog Box of the FootPath

Double-click the icon of the *FootPath* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

Tab Attributes

Tab Attributes

The tab **Attributes** provides the settings, which the object offers.

The tab **Attributes** provides these settings:

[Length \[text box\] - FootPath](#)

[Width \[text box\] - FootPath](#)

The shared properties are described under the [Tab Attributes](#).

Length [text box] - FootPath

Type the **Length** of the *FootPath* into the text box. After you inserted the *FootPath*, *Plant Simulation* shows its length here.

Length	7	 m
Width	0.9	 m

Remarks

The *FootPath* can accommodate any number of *Workers*, as they themselves do not have a length. A *Worker* enters the *FootPath* at position 0 and exits it after covering the **Length** you type in.

You can change the length of the graphic and the anchor points of the *Footpath* by clicking [Show](#)

Manipulators  on the **Edit** ribbon tab or by clicking **M** on the keyboard.

SimTalk

Length [SimTalk] - FootPath

Width [text box] - FootPath

Type the **Width** of the *FootPath* into the text box. After you inserted the *FootPath*, *Plant Simulation* shows its width here.

Length	7	 m
Width	0.9	 m

SimTalk

Width [SimTalk] - FootPath

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The **View Menu** provides commands to access its functions.

[Refresh \[on View menu\]](#)

[Show Attributes and Methods \[on View menu\]](#)

[Help Menu \[in object dialogs\]](#)

SimTalk

[updateDialog \[SimTalk\]](#)

See also

[View Menu \[general description\]](#)

Tools Menu

The **Tools Menu** provides these menu commands:

Edit Controls

Edit Observers

Help Menu

The commands are described under the [Help Menu](#).

Methods of the Footpath

Methods of the FootPath

The *FootPath* provides the [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			((Contents:table)) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			((Index:integer:=1)) -> object
MUPart	VOID			((Index:integer:=1)) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).

An example of the **Syntax** line of the individual methods might look like this:

```
<Path>.openDialog([CallOpenControl:boolean:=false]) → boolean
```